

JAVASCRIPT

1. Introduction to JavaScript

- **What is JavaScript?**
 - History and evolution
 - Client-side vs. server-side scripting
- **Setting up your environment**
 - Browsers and Developer Tools
 - Installing a code editor (e.g., VS Code)
 - Basic debugging in Developer Tools

2. JavaScript Basics

- **Syntax and Fundamentals**
 - Variables (`var`, `let`, `const`)
 - Data types (string, number, boolean, null, undefined, symbol, bigint)
 - Operators (arithmetic, assignment, comparison, logical)
- **Control Structures**
 - Conditionals (`if`, `else`, `else if`, `switch`)
 - Loops (`for`, `while`, `do...while`, `for...of`, `for...in`)
- **Functions**
 - Function declarations and expressions
 - Arrow functions
 - Parameters and return values
 - Scope and closures

3. DOM Manipulation

- **Document Object Model (DOM)**
 - Selecting elements (`getElementById`, `querySelector`, etc.)
 - Modifying HTML and CSS (`innerHTML`, `style`, `classList`)
 - Event handling (`addEventListener`, event types)
- **Forms and Input**
 - Form validation
 - Event listeners for forms
 - Dynamic form inputs

4. Intermediate Concepts

- **Arrays and Objects**
 - Creating and manipulating arrays
 - Array methods (`map`, `filter`, `reduce`, `forEach`)
 - Working with objects and object methods
- **Error Handling**
 - `try...catch`
 - `throw` statements
 - Custom error messages
- **Asynchronous JavaScript**
 - Callbacks
 - Promises (`then`, `catch`, `finally`)
 - `async/await`

5. Advanced Topics

- **ES6+ Features**
 - Destructuring
 - Template literals
 - Default parameters
 - Rest and spread operators
 - Modules (`import` and `export`)
- **Object-Oriented Programming (OOP)**
 - Prototypes and inheritance
 - Classes and constructors
 - Encapsulation, inheritance, and polymorphism
- **Functional Programming**
 - Pure functions
 - Higher-order functions
 - Immutability
 - Currying and partial application

6. Working with APIs

- **AJAX and Fetch API**
 - Making HTTP requests
 - Consuming APIs
 - Error handling in API calls
- **JSON**
 - Parsing and stringifying JSON data
 - Working with API responses

7. Browser-Specific Features

- **LocalStorage, SessionStorage, and Cookies**
- **Geolocation API**
- **Web Workers**
- **Canvas and SVG**

8. Testing and Debugging

- **Debugging Techniques**
 - Using breakpoints
 - The `console` object (`log`, `warn`, `error`, `table`)
- **Testing Frameworks**
 - Unit testing with Jest or Mocha
 - Writing test cases

9. JavaScript Build Tools

- **Introduction to npm and Yarn**
- **Module bundlers**
 - Webpack
 - Parcel
- **Transpilers**
 - Babel

10. Frameworks and Libraries

- **Introduction to Popular Frameworks**
 - React.js
 - Vue.js
 - Angular
- **Utility Libraries**
 - Lodash
 - Moment.js (or alternatives like Day.js)

11. Real-World Applications

- **Building Projects**
 - To-Do List
 - Weather App (using an API)
 - Portfolio Website
 - E-commerce Cart
 - Real-time Chat App
- **Version Control**
 - Git basics

- Hosting projects on GitHub

12. Performance Optimization

- **Optimizing JavaScript**
 - Minification and bundling
 - Lazy loading
 - Debouncing and throttling
- **Web Performance APIs**
 - Timing APIs
 - Analyzing performance in DevTools

13. Advanced Topics (Optional)

- **TypeScript**
 - Introduction and usage with JavaScript
- **Node.js**
 - Setting up a backend
 - Creating REST APIs
- **WebAssembly**
 - Introduction and use c

Jest (Testing-Javascript+Jest)

1. Prerequisites

Before starting Jest, ensure you have a solid understanding of:

1. **JavaScript Basics**
 - ES6+ features: `let/const`, arrow functions, template literals, `async/await`.
 - Functions, arrays, objects, and error handling.
2. **Node.js Basics**
 - Setting up a Node.js environment.
 - Installing npm packages.
3. **Optional**
 - Familiarity with a JavaScript framework/library like React or Vue if testing frontend applications.

2. Introduction to Jest

- **What is Jest?**
 - Overview and features of Jest.
 - Comparison with other testing frameworks (e.g., Mocha, Jasmine).
- **Installing Jest**
 - Setting up Jest in a Node.js project.
 - Running your first Jest test.

3. Core Concepts

- **Test Basics**
 - Writing test cases with `test` and `it`.
 - Using `expect` for assertions.
- **Matchers**
 - Common matchers: `.toBe`, `.toEqual`, `.toContain`.
 - Advanced matchers: `.toHaveBeenCalled`, `.toThrow`.
- **Test Suites**
 - Grouping tests with `describe`.

4. Mocking

- **Manual Mocks**
 - Creating manual mocks for dependencies.
- **Mock Functions**
 - Using `jest.fn()` to create mock functions.
 - Asserting on mock calls with `.mock.calls`.
- **Spies**
 - Using `jest.spyOn()` to monitor function calls.
- **Mock Modules**
 - Automating module mocking with `jest.mock()`.

5. Asynchronous Testing

- **Promises**
 - Testing functions that return promises.
- **Async/Await**
 - Using `async/await` in test cases.

- **Callbacks**
 - Testing functions that use callbacks.

6. Advanced Jest Features

- **Snapshots**
 - Creating and maintaining snapshots with `toMatchSnapshot`.
 - Updating snapshots with `--updateSnapshot`.
- **Timers**
 - Mocking timers with `jest.useFakeTimers()`.
 - Testing delayed code with `jest.runAllTimers()`.
- **Custom Matchers**
 - Extending Jest with custom matchers.
- **Parameterization**
 - Using `test.each` for data-driven tests.

7. Integration Testing

- **Setup and Teardown**
 - Using `beforeAll`, `beforeEach`, `afterAll`, and `afterEach`.
- **Database Mocking**
 - Mocking database calls and external APIs.
- **Testing Frameworks**
 - Combining Jest with supertest for testing REST APIs.

8. Testing Framework Integration

- **React Testing**
 - Setting up Jest with React and React Testing Library.
 - Testing components, hooks, and props.
- **Vue Testing**
 - Integrating Jest with Vue Test Utils.
 - Testing Vue components and directives.
- **Node.js APIs**
 - Writing tests for REST APIs and backend logic.

9. Performance and Debugging

- **Debugging Tests**
 - Using `console.log` and `jest.runOnlyPendingTimers()` for debugging.
 - Running tests in watch mode (`jest --watch`).
- **Performance Optimization**
 - Running tests in parallel with `jest --maxWorkers`.
 - Using `--runInBand` for single-threaded execution in specific cases.

10. CI/CD Integration

- **Running Jest in CI/CD**
 - Configuring Jest for CI/CD pipelines.
 - Generating coverage reports with `--coverage`.

11. Best Practices

- Writing clear and isolated tests.
- Keeping test cases and code coverage balanced.
- Structuring test files and folders.

12. Advanced Topics

- **Jest Configuration**
 - Customizing Jest with `jest.config.js`.
 - Configuring Babel or TypeScript with Jest.
- **Testing Libraries**
 - Using Jest with Enzyme or React Testing Library for React.
- **Mocking Complex Scenarios**
 - Mocking chained functions or complex dependencies.
- **Monorepos**
 - Setting up Jest for projects with multiple packages.